Labor Supply and Saving under Uncertainty*

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Abstract

This paper examines how variations in labor supply can be used to self insure against individual specific risks, and what implications this has for the magnitude of precautionary savings. It is found that labor-supply flexibility tends to increase precautionary savings when future wages are uncertain. Maybe surprisingly, the effects of precautionary saving can be significantly underestimated if labor supply is assumed fixed when it is not. This is the case even if utility is separable in consumption and leisure. The precautionary effects on leisure are similar to those on consumption. Agents work more today and less tomorrow if risk increases.

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1 Introduction

How can variations in labor supply be used to self insure against individual-specific risks? In spite of the significant amount of work which has been devoted to both theoretical and quantitative studies of precautionary saving, this question seems to be unanswered.

Pratt (1964) showed that for an agent with utility function $u(c)$, where c is consumption, $-u_{cc}/u_c$ is a good measure of his absolute degree of risk aversion. While risk aversion is a concept closely related to an agent's utility, prudence and precautionary saving relate to the agent's behavior in the presence of uncertainty. Leland (1968) and Sandmo (1970) first formalized this concept and showed that a positive third derivative of the utility function is crucial for obtaining precautionary saving. Kimball (1990) paralleled Pratt's analysis and showed that a good measure of the absolute degree of prudence is $-u_{ccc}/u_{cc}$.

In this paper, I examine the precautionary behavior of agents who can choose their labor supply in each period. First, I look at the concepts of prudence and precautionary behavior in this setting, and relate them to Kimball's study. The theoretical implications of precautionary behavior are then analyzed in a simple two-period model. In particular, I examine how agents use variations in labor supply to self-insure against risk, and what implications this has for the magnitude of precautionary savings.

The standard measure of prudence, as defined by Kimball, does not always correctly measure precautionary strength in this setting. Kimball defines prudence from an equivalent variation premium on the stochastic variable. This premium will be related to the stochastic variable and need not indicate how decision variables respond to uncertainty. For example, the standard definition of prudence works well when studying precautionary saving only if labor supply is fixed and saving and shocks are measured in the same units. In the next section I look into this problem and suggest how to measure the precautionary strength when labor supply is flexible.

Eaton and Rosen (1980) showed that labor supply can increase in response to increased wage uncertainty if risk aversion is sufficiently high. I show here that when allowing for an interaction between labor supply and saving decisions, increased wage uncertainty unambiguously reduces future labor supply. This finding is intuitive. Just as increased uncertainty tends to increase future consumption, it tends to increase future leisure. Using a somewhat different setup, Marcet et al. (2002) come to the same conclusion. In another related paper, Bodie, Merton, and Samuelson (1992) analyzed how labor-supply flexibility influences investors' portfolio decisions. One of their findings was that greater flexibility induces more risk taking. Therefore, one might expect that greater labor-supply flexibility also would make agents less prudent. However, this will not always be the case.

The analytical and numerical investigations in this paper show that labor-supply flexibility raises precautionary savings when wages are stochastic. Of course, flexibility is always good for the agents, so even if their precautionary savings increase, expected utility is higher with flexible labor supply. With fixed labor supply, all effects of a negative shock must be absorbed by consumption. With flexible labor supply, hours worked can be adjusted to alleviate the effect of the shock. By the same argument we note that a certain amount of savings is less costly for agents when labor supply is flexible. Therefore, agents with flexible labor supply are willing to expose themselves to more risk but they can more easily save to self insure against uncertainty.

This last point, that flexibility makes saving easier, resembles the Le Chatelier-Samuelson principle (see Samuelson, 1972). The contents of this principle is that the elasticity of demand of one variable is greater when other variables are allowed to adjust to price changes than when other variables are held fixed. In the present case, the amount of uncertainty is related to the

value of saving. Here then, saving will increase more in response to increased uncertainty if labor supply is flexible, provided that this effect dominates the effect on risk tolerance.

The standard measure of prudence is closely related to risk aversion. Kimball and Weil (2003) broke that link, and showed that both high risk aversion and high intertemporal elasticity of substitution tend to imply much prudence. The present paper illustrates this point. If an agent has decreasing absolute risk aversion, he can insure against wage fluctuations by bringing much wealth into the risky period, and if the intertemporal elasticity of substitution is high, it is less costly to shift wealth across periods.

The present paper uses an unrealistically simple model to illustrate the mechanisms behind precautionary saving and thus leaves a number of important questions unanswered. Do the results extend to multi-period models? Is precautionary saving of quantitative significance? To some extent, these questions have been addressed by recent research. Whether the results apply to multi-period dynamic general equilibrium models is not clear. Huggett and Ospina (2001) argue that the existence of precautionary saving need not depend on the properties of the utility function in such models. Their finding thus indicates that the results do not extend to multi-period models. But Hugget and Ospina's findings have to be interpreted carefully. First, they do not say that the properties of the utility function are unimportant for the *magnitude* of savings. Second, their results only apply to economies with potentially binding borrowing constraints. In models with no such constraints, for example Wang (2003), the measure of prudence does matter. Third, it is difficult to separate precautionary saving from life-cycle savings in such models. In the setting studied by Hugget and Ospina, more uncertainty implies more income volatility. And this volatility in combination with liquidity constraints can affect life-cycle savings even if income is perfectly predictable.

The quantitative importance of precautionary savings has also been examined. Aiyagari (1994) found precautionary savings to be modest in a dynamic general equilibrium model with fixed labor supply, at least for his preferred parameterizations of income uncertainty. But recent evidence (for example Storesletten et al., 2003) indicate that income processes are substantially more volatile and persistent than assumed in the early quantitative models. The results in the present paper further suggest that the amount of precautionary savings can be underestimated if labor-supply flexibility is ignored.

Recent work by Low (2002) and Marcet et al. (2002) examine how labor-supply decisions, savings, and uncertainty interact in dynamic general equilibrium models. Low's (2002) paper is particularly interesting since it illustrates many of the mechanisms that I examine theoretically. He assumes Cobb-Douglas utility and solves numerically a life-cycle model with wage uncertainty for different values of the intertemporal elasticity. He finds that uncertainty raises labor supply of young agents. This is consistent with my findings in Section 4: more future uncertainty implies more labor supply today. Low further finds that there is more savings when labor supply is flexible rather than fixed, and he finds a U-shaped relationship between total savings and the intertemporal elasticity when labor supply is flexible. When labor supply is fixed, he finds a negative relationship between savings and the elasticity. All this is consistent with my theoretical analysis based on Cobb-Douglas utility (Section 3.2). I interpret this as a strong indication that the properties of the utility function are indeed important for understanding precautionary behavior in dynamic general equilibrium models.

The remainder of the paper is structured as follows. In Section 2, I suggest a way of measuring the strength of precautionary motives when labor supply is endogenous. Thereafter, in Section 3, I use the measure of precautionary strength to examine how labor-supply flexibility affects precautionary savings in a two-period economy. I show that agents with flexible labor supply

typically choose to save more. In Section 4, I consider how uncertainty affects incentives to work. I show that more wage uncertainty unambiguously has a positive effect on labor supply today but a negative effect on future labor supply. Section 5 concludes.

2 Prudence and precautionary behavior

Let us first consider a general framework with notation similar to Kimball (1990). I suggest how to measure of precautionary strength in this framework, and I demonstrate that Kimball's definition of prudence does not generally result in a measure that is directly related to the magnitude of precautionary reactions.

Let θ denote a stochastic variable and denote its mean by $\bar{\theta}$ such that $\theta = \bar{\theta} + \tilde{\theta}$, where $\tilde{\theta}$ is the stochastic component with variance σ_θ^2. Further, let δ denote the decision variable, and let $v(\delta, \theta)$ be the agent's indirect utility function. Finally, let E_θ denote expectations over θ.

The agent chooses δ to solve

$$\max_\delta \mathrm{E}_\theta V(\delta, \theta).$$

The first order condition is then

$$\mathrm{E}_\theta V_\delta(\delta, \theta) = 0.$$

To measure how much the decision variable δ changes in response to uncertainty, consider first the decision under certainty. The first-order condition is then $V_\delta\left(\bar{\delta}, \bar{\theta}\right) = 0$ where $\bar{\delta}$ is the optimal decision. The relevant precautionary premium is therefore $\psi = \delta - \bar{\delta}$, i.e. the difference between the agent's choice under certainty and the choice under uncertainty. By combining the first-order conditions, we have

$$\mathrm{E}_\theta V_\delta(\delta, \theta) = V_\delta\left(\delta - \psi, \bar{\theta}\right). \tag{1}$$

A second-order expansion of this around δ and $\bar{\theta}$ gives

$$\psi = -\frac{V_{\delta\theta\theta}\left(\delta, \bar{\theta}\right)}{V_{\delta\delta}\left(\delta, \bar{\theta}\right)} \frac{\sigma_\theta^2}{2}$$

and I suggest that the precautionary strength is measured by

$$\eta \equiv -\frac{V_{\delta\theta\theta}}{V_{\delta\delta}}. \tag{2}$$

This shows that the amount of precautionary savings is determined by the convexity of marginal utility in θ ($V_{\delta\theta\theta}$) and by the response of marginal utility to changes in the decision variable ($V_{\delta\delta}$).

The definition of precautionary strength I have proposed deviates slightly from Kimball's (1990) definition of prudence. Kimball measures prudence by the premium, $\hat{\psi}$, which fulfills

$$\mathrm{E}_\theta V_\delta(\delta, \theta) = V_\delta\left(\delta, \bar{\theta} - \hat{\psi}\right). \tag{3}$$

In words, this definition says that if the agent gets $\bar{\theta} - \hat{\psi}$ for sure, instead of the uncertain θ, his decision will be the same. A second-order expansion of (3) gives

$$\hat{\psi} = -\frac{V_{\delta\theta\theta}\left(\delta, \bar{\theta}\right)}{V_{\delta\theta}\left(\delta, \bar{\theta}\right)} \frac{\sigma_\theta^2}{2}.$$

Kimball consequently defines prudence as

$$\hat{\eta} = -\frac{V_{\delta\theta\theta}}{V_{\delta\theta}}.$$

A problem with Kimball's definition of prudence is that it need not measure the magnitude of the agent's reaction to uncertainty. To understand this problem, note that the precautionary premium $\hat{\psi}$ is a measure related to the stochastic variable θ rather than to the decision variable δ. Let for example ξ be a linear transformation of θ, $\xi = \theta/a$ and let X denote the value function expressed in terms of ξ so that $X(\delta, \xi) \equiv V(\delta, a\xi)$. This implies that the precautionary strength for the value function X is

$$\hat{\psi}^X = -\frac{X_{\delta\xi\xi}}{X_{\delta\xi}}\frac{\sigma_\xi^2}{2} = \frac{\hat{\psi}}{a}.$$

By measuring the uncertain variable in another unit, the optimal decision δ does not change, and the measure of precautionary strength should be unaffected. One can easily verify that the definition proposed in (1) does not depend on such transformations of the stochastic variable.

2.1 Example

Let us solve for precautionary savings in a simple two-period model. Assume that the agent has a time-separable utility function, abstract from discounting and assume zero interest rates. Further, assume that the first-period and second-period utility functions are identical and that the agent has no initial financial wealth. Let s be savings and ε a second-period shock with mean zero and variance σ^2, and define $\bar{v}(s)$ and $v(s, \varepsilon)$ as the indirect first-period and second-period utility functions respectively. The agent's indirect life-time utility is then defined as

$$V(s, \varepsilon) \equiv \bar{v}(s) + v(s, \varepsilon).$$

The agent chooses savings in the first period, s, to maximize his expected indirect utility $\mathrm{E}_\varepsilon V(s, \varepsilon)$. The first-order condition is

$$\bar{v}_s(s) + \mathrm{E}_\varepsilon v_s(s, \varepsilon) = 0. \tag{4}$$

Expand $\bar{v}_s$ and v_s around $s = 0$ and $\varepsilon = 0$,and ignore high-order terms, to get

$$\bar{v}_s(s) = \bar{v}_s(0) + \bar{v}_{ss}(0)\, s,$$

and

$$\begin{aligned}
\mathrm{E}_\varepsilon v_s(s, \varepsilon) &= \mathrm{E}_\varepsilon \left[v_s(0,0) + v_{ss}(0,0)\, s + v_{s\varepsilon}(0,0)\, \varepsilon + \frac{1}{2} v_{s\varepsilon\varepsilon}(0,0)\, \varepsilon^2 + v_{ss\varepsilon}(0,0)\, s\varepsilon \right] \\
&= v_s(0,0) + v_{ss}(0,0)\, s + \frac{1}{2} v_{s\varepsilon\varepsilon}(0,0)\, \sigma^2.
\end{aligned}$$

Assume now that $s = 0$ solves (4) when $\sigma^2 = 0$. Then $v_s(0,0) = -\bar{v}_s(0)$, and $v_{ss}(0,0) = \bar{v}_{ss}(0)$. Hence, if (4) is fulfilled we have that

$$0 = [-v_s(0,0) + v_{ss}(0,0)\, s] + \left[v_s(0,0) + v_{ss}(0,0)\, s + \frac{1}{2} v_{s\varepsilon\varepsilon}(0,0)\, \sigma^2 \right].$$

We get the implied amount of precautionary savings in response to a small uncertainty by rearranging,

$$s = -\frac{v_{s\varepsilon\varepsilon}}{v_{ss}}\frac{\sigma^2}{4}, \tag{5}$$

With the previous notation, ε is the uncertain variable θ, and s is the decision variable δ. The measure of precautionary strength proposed in (1) is thus consistent with what was found in this example (compare equations [2] and [5]), while the standard definition in (3) is not. The two definitions result in equivalent measures of precautionary strength if the choice variable and the stochastic variable enter additively in the value function and are measured in the same units, as for example in the cases analyzed by Kimball (1990). When analyzing how labor-supply flexibility affects precautionary savings, it will however be necessary to use the modified measure. Numerical examples in the next section (see Table I) show that the standard measure of prudence may be a misleading indicator of precautionary savings.

3 Labor supply and precautionary savings

In this section, I apply the measure of precautionary strength in a two-period framework where agents derive utility from consumption and leisure. I use these measures on explicit utility functions and examine how wage uncertainty and labor-supply flexibility influences precautionary saving.

The agent is assumed to have a time separable utility function, $u(c, l)$, where c is his consumption of goods, and l is his consumption of leisure. I restrict the study to utility functions which are strictly concave, i.e. $u_{cc}, u_{ll} < 0$, $u_{cc}u_{ll} - u_{cl}^2 > 0$, and 'precautionary', i.e. $u_c, u_l, u_{ccc}, u_{lll} > 0$. Further, the agent is assumed to have one unit of time to dispose of, and throughout the analysis I assume interior solutions for the leisure choice.

The agent lives for two periods and chooses consumption and labor supply in each period. The first period wage rate is certain, $w_1 = w$, while the second-period wage w_2 is uncertain. The budget constraint is thus

$$c_1 = (1 - l_1)\, w - s, \tag{6}$$

$$c_2 = (1 - l_2)\, w_2 + s, \tag{7}$$

where s is savings. Let ε denote a second-period wage shock with zero mean, and assume that $w_2 = w + \varepsilon$.

The agent chooses consumption and labor supply to maximize $u(c_1, l_1) + \mathrm{E}_\varepsilon u(c_2, l_2)$. Given the time separable utility, the leisure choice in each period is a function of contemporaneous consumption and wages,

$$u_l(c_t, l_t) = w_t u_c(c_t, l_t). \tag{8}$$

This, together with (6) and (7) defines the leisure choices as functions of savings and the shock, $l_1 \equiv L_1(s)$, and $l_2 \equiv L_2(s, \varepsilon)$. The indirect first-period utility function is then

$$\bar{v}(s) = u\left[(1 - L_1(s))\, w - s, L_1(s)\right],$$

and the indirect second-period utility is

$$v(s, \varepsilon) = u\left[(1 - L_2(s, \varepsilon))\, w_2 + s, L_2(s, \varepsilon)\right].$$

For future reference, note that by applying the Envelope theorem on this last equation, we get

$$v_s(s,\varepsilon) = u_c. \tag{9}$$

Define also

$$L_\varepsilon \equiv \frac{\partial L_2(s,\varepsilon)}{\partial \varepsilon}, \text{ and } L_s \equiv \frac{\partial L_2(s,\varepsilon)}{\partial s}.$$

The parameters in the utility function are often chosen so that either the intertemporal elasticity of substitution or the degree of risk aversion get a plausible value. The same parameter values may however result in different intertemporal elasticities and degrees of risk aversions when labor supply is flexible rather than fixed. In general, it is therefore not straightforward to compare utility functions under these different assumptions. The utility functions considered below do not suffer from these problems, however. If utility is additively separable in consumption and leisure, the parameter choices will be indifferent to the assumptions about labor-supply flexibility. In the appendix, I also demonstrate that risk aversion is not affected by labor-supply flexibility when utility has the Cobb-Douglas form. Therefore, if any of these utility functions is calibrated to match a certain degree of risk aversion, the calibration will be the same when labor supply is assumed to be endogenous as when labor supply is assumed to be fixed.

Consider first the measure of precautionary strength when labor supply is not a choice variable. Let l denote the level that would be chosen under certainty, i.e. $\bar{l} = L_1(0) = L_2(0,0)$. Since labor supply is fixed, $L_\varepsilon = L_s = 0$ by assumption, and we obtain the measure of prudence

$$\eta^{\text{fix}} = -\frac{v_{s\varepsilon\varepsilon}}{v_{ss}} = -\frac{\left(1-\bar{l}\right)^2 u_{ccc}}{u_{cc}}. \tag{10}$$

3.1 Separable utility

Let us instead assume that the agent can choose labor input and that the utility function is separable in consumption and leisure so that $u_{cl} = 0$. The intratemporal equilibrium condition (8) holds for all ε and s, so

$$L_\varepsilon = \frac{(1-L_2)\, w_2 u_{cc} + u_c}{u_{ll} + w_2^2 u_{cc}},$$

and

$$L_s = \frac{w_2 u_{cc}}{u_{ll} + w_2^2 u_{cc}}. \tag{11}$$

The precautionary strength is then

$$\eta^{\text{flex}} = -\frac{v_{s\varepsilon\varepsilon}}{v_{ss}} = \frac{-\left(1-L-L_\varepsilon w\right)^2 u_{ccc}}{\left(1-L_s w\right) u_{cc}} + \frac{2L_\varepsilon + L_{\varepsilon\varepsilon} w}{1-L_s w}. \tag{12}$$

There are several mechanisms at work here. Labor supply changes due to a substitution effect, an income effect and a wealth effect. The substitution and income effects are due to fluctuations in the wage rate which changes the price on leisure relative to consumption. The wealth effect arises since prudent consumption behavior changes saving. Moreover, second-period labor supply changes in response to wage shocks, so there will be a precautionary effect on leisure when $u_{lll} > 0$. Hence, first-period leisure will tend to fall and this tends to increase first-period saving. However, for some utility functions, a reduction in leisure will tend to increase consumption of goods and hence reduce saving.

The above expressions do not provide a clear answer to whether saving is higher or lower when labor supply is flexible. To address that question, consider utility functions that are logarithmic in consumption so that income and substitution effects cancel,

$$u^A(c,l) = \ln c + r(l)$$

where r is an increasing and strictly concave function. With these utility functions, savings is higher with flexible labor supply. To see this, note that when $s = 0$ leisure and consumption are determined by $l = 1 - c/w$ and $r'(l) = w/c$, where the first equation is the budget constraint and the second equation is the first-order condition for an optimum. These equations imply that c/w must be constant when w changes. Consequently $L_\varepsilon(0,\varepsilon) = 0$. Since $0 < (1 - L_s w) < 1$, equations (12) and (10) imply that $\eta^{\text{flex}} > \eta^{\text{fix}}$, i.e. there is more precautionary saving with labor-supply flexibility.

Does this also mean that more flexibility implies more precautionary savings? To answer this question assume that

$$r(l) = \frac{b\left(l^{1-1/\gamma} - 1\right)}{1 - 1/\gamma}$$

where γ is the intertemporal elasticity of leisure, $b = \bar{l}^{1/\gamma}/\left(1 - \bar{l}\right)$ is a constant, and $\bar{l}$ denotes the level of leisure agents would choose under certainty. Part (a) of Proposition 1 establishes that precautionary savings increase as leisure becomes more elastic in the utility function. Part (b) shows that when leisure becomes totally inelastic, agents choose the same savings as when labor supply is fixed by regulation. The proof of this proposition is in the appendix.

Proposition 1

(a) $\frac{\partial \eta^{\text{flex}}}{\partial \gamma} > 0$

(b) $\lim_{\gamma \to 0} \eta^{\text{flex}} = \eta^{\text{fix}}$.

3.2 Cobb-Douglas utility

Assume now that utility is of the Cobb-Douglas type,

$$u^C(c,l) = \frac{\left(c^\alpha l^{1-\alpha}\right)^{1-1/\gamma} - 1}{1 - 1/\gamma}.$$

where $\gamma > 0$ is the intertemporal elasticity of substitution and $0 < \alpha < 1$. We can again show that labor-supply flexibility implies more precautionary saving, i.e. $\eta^{\text{flex}} > \eta^{\text{fix}}$.

When labor supply is fixed,

$$-\frac{u_{ccc}}{u_{cc}} = \frac{2 - \alpha(1 - 1/\gamma)}{\mathrm{E}_\varepsilon c_2} = \frac{2 - \alpha(1 - 1/\gamma)}{\alpha w}.$$

So, from (10)

$$\eta^{\text{fix}} = \frac{\alpha\left[2 - \alpha(1 - 1/\gamma)\right]}{w}.$$

When labor supply is a choice variable, we get the equilibrium condition

$$l_2 = \frac{1-\alpha}{\alpha}\frac{c_2}{w_2}, \tag{13}$$

so $c_2 = \alpha (w + \varepsilon + s)$. The indirect second-period utility function is then

$$v(s, \varepsilon) = \frac{(w + \varepsilon + s)^{1-1/\gamma}}{1 - 1/\gamma} (w + \varepsilon)^{-(1-\alpha)(1-1/\gamma)} K, \tag{14}$$

where K is a constant. Differentiate (14) to get

$$\eta^{\text{flex}} = -\frac{v_{s\varepsilon\varepsilon}(0,0)}{v_{ss}(0,0)} = \frac{\gamma [1 - \alpha (1 - 1/\gamma)] [2 - \alpha (1 - 1/\gamma)]}{w}.$$

We know that $1 - \alpha > 0$ and $\gamma > 0$ so $\gamma [1 - \alpha (1 - 1/\gamma)] > \alpha$ which shows that $\eta^{\text{flex}} > \eta^{\text{fix}}$.

Figure 1, plots the measure of precautionary strength with fixed and flexible labor supply against the leisure-elasticity. The figure shows that the precautionary strength is higher when labor supply is flexible than when it is fixed, which is what we just demonstrated analytically. More interestingly, the figure displays a U-shaped relationship between the intertemporal elasticity and the precautionary strength when labor supply is flexible. This is in accordance with Kimball and Weil's (2003) finding that both high intertemporal elasticity of substitution and high risk aversion imply much precautionary saving.[1] When the intertemporal elasticity is low, a small increase in the elasticity leads to a substantial fall in risk aversion.[2] The large reduction in risk aversion then dominates over the small increase in elasticity. The degree of risk aversion does not fall to zero as the intertemporal elasticity of substitution γ increases to infinity. If γ is high, a further increases in the intertemporal elasticity implies a minor reduction in risk aversion. When labor supply is flexible, the effect of higher intertemporal elasticity then dominates over the fall in risk aversion and precautionary motives increase.

3.3 Numerical examples

When looking at Figure 1, it should be noted that the relevance of η as a measure of precautionary saving is derived under the assumption that saving is small. Therefore, as γ approaches infinity and precautionary savings increases, it is possible that η loses its connection to the amount of savings.

To evaluate the validity of the precautionary measures for non-negligible risks, I have calculated the savings predicted by these measures in conjunction with equation (5). For various amounts of wage uncertainty, I have also solved numerically for savings directly from the Euler equation (4). The results are shown in Table 1. It is clear that the measure of precautionary strength is strongly related to the actual savings chosen by agents. The level of saving is well predicted when the standard deviation of wages is ten percent. When the standard deviation is twice as high, predictions are still roughly accurate, but saving is consistently underestimated. Note also that savings is lower when labor supply is fixed rather than flexible, as we also showed theoretically for both utility functions. Consistent with Proposition 1, we also see that the difference between fixed and flexible labor supply increases as the intertemporal elasticity increases.

Table 1 also reports the savings implied if the standard measure of prudence, $\hat{\eta}$, is used in (5).[3] The table clearly shows that this measure of prudence is misleading when labor supply is a

[1] If risk aversion is high and if the agent has decreasing absolute risk aversion, precautionary behavior reduces the utility cost of uncertainty. If intertemporal elasticity is high, the utility cost of precautionary behavior that reallocates resources between periods is low.

[2] See the appendix for calculations of the absolute degree of risk aversion displayed in Figure 1.

[3] This measure is $\hat{s}^{\text{flex}} = (1 - \bar{l}) \sigma^2 / (2c)$ with separable utility and $\hat{s}^{\text{flex}} = [2 - \alpha (1 - 1/\gamma)] \sigma^2 / (4w)$ with Cobb-Douglas utility.

choice variable, and even if utility is separable in consumption and leisure. The two definitions of precautionary strength are identical, $\hat{s}^{\text{fix}} = s^{\text{fix}}$, when labor supply is not a choice variable. Using the standard measure of prudence is then not a problem.

4 Precautionary labor supply

The previous analysis has indicated that more labor-supply flexibility can increase the amount of precautionary savings. However, the exact ways in which agents use variations in labor supply to insure against shocks are diffuse since labor supply reacts to realized wage shocks. To isolate the effects from uncertainty to the level of labor supply, let us assume that second-period labor supply must be chosen before uncertainty is resolved. Eaton and Rosen (1980) use the same timing, but they do not allow for precautionary saving. In contrast to Eaton and Rosen's analysis, we will see that increased wage uncertainty unambiguously reduces second-period labor supply.

The second period wage is again uncertain, $w_2 = w + \varepsilon$, and we define savings as

$$s = (1 - l_1) w_1 - c_1.$$

Consequently, agents maximize

$$u(c_1, l_1) + \mathrm{E}_\varepsilon u(c_2, l_2),$$

subject to

$$c_2 = (1 - l_2) w_2 + s.$$

We want to see how s and l_2 respond to increased uncertainty.[4] To do this, rewrite first period consumption and leisure as functions of savings, and let σ denote the uncertainty in w_2.[5] Then define

$$f(s) \equiv u_c [c_1(s), l_1(s)],$$

$$g(s, l_2; \sigma) \equiv \mathrm{E}_\varepsilon u_c (c_2, l_2),$$

and

$$h(s, l_2; \sigma) \equiv \mathrm{E}_\varepsilon [w_2 u_c (c_2, l_2) - u_l (c_2, l_2)].$$

The relevant first order conditions are then

$$f(s) = g(s, l_2; \sigma) \tag{15}$$

and

$$h(s, l_2; \sigma) = 0. \tag{16}$$

Proposition 2 states that agents choose more second period leisure and first period saving when wage uncertainty is increased.

Proposition 2 *Assume that the utility function is additively separable in consumption and leisure. Then, for small σ,*

$$\frac{dl_2}{d\sigma} > 0,$$

[4] It is clear that both c_1 and l_1 fall when savings increase.

[5] More precisely, σ is defined such that a higher σ implies more risk in the Rotschild - Stiglitz sense. So, if two stochastic variables have different σ, the variable with the highest σ can be constructed by adding a mean preserving spread to the other.

and

$$\frac{ds}{d\sigma} > 0.$$

Proof. See the Appendix.

By reducing second-period labor supply, the agent avoids some risk, and for decreasing absolute risk aversion, increased saving also reduces risk. The agent can compensate the second-period income loss (due to less work) by working more and consuming less in the first period. The precautionary effects on leisure are thus similar to those on consumption: future uncertainty reduces current consumption and leisure and increases future consumption and leisure.

5 Conclusions

This paper has considered precautionary behavior of agents with flexible labor supply in a simple two-period model. The main insights of the paper are that labor-supply flexibility tends to increase savings when future wages are uncertain and that future wage uncertainty tends to raise current labor supply. The first result may be surprising since more flexibility increases risk tolerance. Working hard in periods with low wages is, however, a bad strategy. In a previous version of the paper I also considered the effects of wealth shocks. Subject to such shocks, labor-supply flexibility implies less precautionary savings.

Appendix A: Proofs and calculations

A.1 Risk aversion for Cobb-Douglas utility

We use (14), to find the degree of absolute risk aversion, evaluated at $s = 0$, for agents with flexible labor supply,[6]

$$r^{\text{flex}} = -\frac{v_{\varepsilon\varepsilon}}{v_s} = \frac{\alpha^2/\gamma^{\text{flex}} + \alpha(1-\alpha)}{w}.$$

An agent with labor supply fixed at $\bar{l} = 1 - \alpha$ consumes $c_2 = \alpha w_2 + s$,and his degree of risk aversion is

$$r^{\text{fix}} = -\frac{\left(1-\bar{l}\right)^2 u_{cc}}{u_c} = \frac{\alpha^2/\gamma^{\text{fix}} + \alpha(1-\alpha)}{w}.$$

We see that the two agents have the same degree of risk aversion if they have the same γ.

A.2 Proof of Proposition 1

Note that $L_\varepsilon = 0$ as argued in the text. Using this fact in (12) we get

$$\eta = \frac{-(1-l)^2 u_{ccc}}{(1-L_s w)\, u_{cc}} = \frac{K_1}{1-L_s w} \tag{A.1}$$

where $K_1 \equiv 2(1-l)^2/c > 0$. In these equations, only L_s depends on γ. Hence

$$\frac{\partial \eta}{\partial \gamma} = \frac{wK_1}{(1-L_s w)^2}\frac{\partial L_s}{\partial \gamma}$$

[6] This is the risk attitude to what Drèze and Modigliani (1972) call "timeless" uncertainty, i.e. the risk agents face after having chosen first period saving. The expression is evaluated at $s = 0$.

and it is clear that

$$\text{sign}\left(\frac{\partial \eta}{\partial \gamma}\right) = \text{sign}\left(\frac{\partial L_s}{\partial \gamma}\right).$$

The derivative of second period leisure with respect to saving is given by (11), $L_s = wu_{cc}/\left(u_{ll} + w^2 u_{cc}\right)$. Consequently

$$\begin{aligned} \frac{\partial L_s}{\partial \gamma} &= -\frac{wu_{cc}}{\left(u_{ll} + w^2 u_{cc}\right)^2} \frac{\partial u_{ll}}{\partial \gamma} \\ &= K_2 \frac{\partial \left(-\frac{1}{\gamma} b l^{-1/\gamma - 1}\right)}{\partial \gamma} \end{aligned}$$

where $K_2 \equiv -wu_{cc}/\left(u_{ll} + w^2 u_{cc}\right)^2 > 0$. The coefficient b is defined as

$$b = \frac{\bar{l}^{1/\gamma}}{1 - \bar{l}},$$

and since $\bar{l} = l|_{s=0, w_2 = w}$ we get

$$\frac{\partial \left(-\frac{1}{\gamma} b l^{-1/\gamma - 1}\right)}{\partial \gamma} = \frac{\partial \left[-\frac{1}{\gamma} l^{-1} (1 - l)^{-1}\right]}{\partial \gamma} = \frac{1}{l(1 - l)} > 0.$$

This shows that

$$\frac{\partial \eta}{\partial \gamma} > 0.$$

which concludes the proof of part (a).

To see that $\lim_{\gamma \to 0} \eta^{\text{flex}} = \eta^{\text{fix}}$, first note that equation (10) here yields

$$\eta^{\text{fix}} = \frac{2(1 - l)^2}{c}$$

which implies that η^{fix} is invariant to γ. From (A.1) we get

$$\eta^{\text{flex}} = \frac{K_1}{1 - L_s(\gamma) w},$$

and (11) gives

$$L_s = \frac{\gamma w / c^2}{\frac{1}{l(1-l)} + \gamma w^2 / c^2}.$$

Since $0 < l < 1$, we see that $\lim_{\gamma \to 0} L_s = 0$. This verifies the Proposition,

$$\lim_{\gamma \to 0} \eta^{\text{flex}} = K_1 = \eta^{\text{fix}}.$$

■

A.3 Proof of Proposition 2

Totally differentiate (15) and (16) to get

$$\frac{dl_2}{d\sigma} = -\frac{g_\sigma h_s + h_\sigma (f_s - g_s)}{(f_s - g_s) h_l + g_l h_s}, \tag{A.2}$$

and

$$\frac{ds}{d\sigma} = \frac{g_\sigma h_l - g_l h_\sigma}{(f_s - g_s) h_l + g_l h_s}. \tag{A.3}$$

We know that $f_s > 0$, $g_s < 0$, $g_l > 0$, $h_s < 0$, and $h_l > 0$. Since we have assumed that $u_{ccc} > 0$, we see that $u_c(c_2, l_2)$ is convex in w_2, and hence that $g_\sigma > 0$, where g_σ is the derivative of g with respect to its third argument. Uncertainty has unclear effects on h, however. If

$$(1 - l_2)\left[w_2 (1 - l_2) u_{ccc} + 2u_{cc} - (1 - l_2) u_{lcc}\right]$$

is positive, then $w_2 u_c - u_l$ is convex in w_2, and if the expression is negative, $w_2 u_c - u_l$ is concave.

For additively separable utility and small s and σ, the denominator in (A.2) and (A.3) is positive since then

$$\begin{aligned} f_s &= -g_s \\ g_l &= -w g_s \\ h_s &= -g_l \\ h_l &= w g_l - u_{ll}(l_2), \end{aligned}$$

that is,

$$(f_s - g_s) h_l + g_l h_s = 2g_s \left(w^2 g_s + u_{ll}\right) - w^2 g_s^2 = w^2 g_s^2 + 2g_s u_{ll} > 0.$$

It remains to show that $g_\sigma h_l - g_l h_\sigma > 0$. Expanding g and h around w, we get for small σ

$$g_\sigma = \chi_1 u_{ccc}$$

and

$$h_\sigma = \chi_1 w u_{ccc} + \chi_2 u_{cc}$$

where $\chi_1 > 0$ and $\chi_2 > 0$. Hence

$$\begin{aligned} g_\sigma h_l - g_l h_\sigma &= \chi_1 u_{ccc} (w g_l - u_{ll}) - g_l (\chi_1 w u_{ccc} + \chi_2 u_{cc}) \\ &= -\chi_1 u_{ccc} u_{ll} - g_l \chi_2 u_{cc} > 0. \end{aligned}$$

This shows that $\frac{ds}{d\sigma} > 0$. It then follows immediately that also $\frac{dl_2}{d\sigma} > 0$. ■

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Table 1

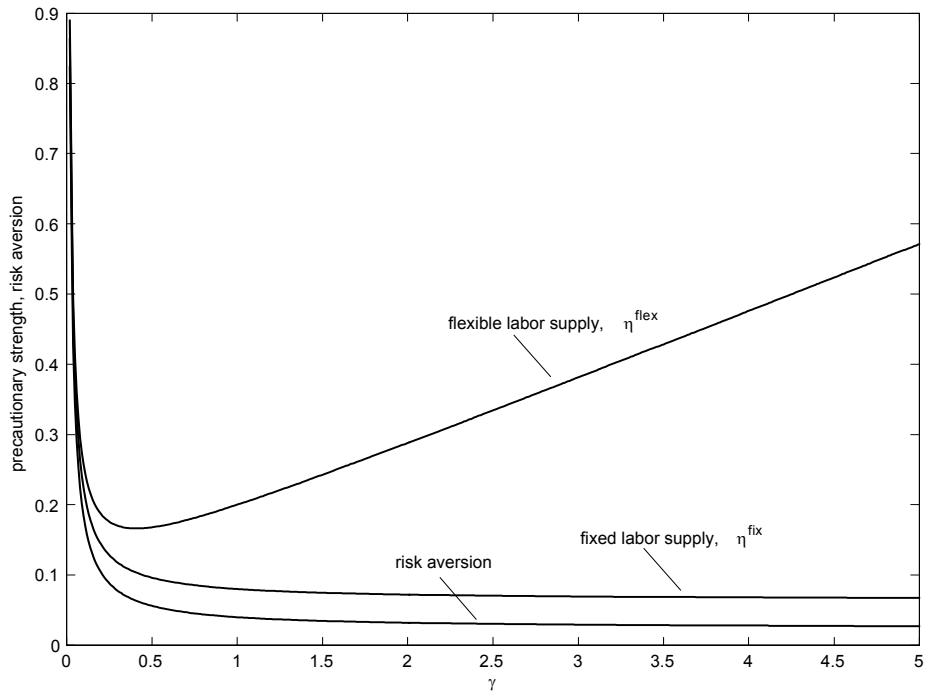
Predictive power of precautionary measure when uncertainty is not negligible

	flexible labor supply						fixed labor supply			
	$\sigma = 1$			$\sigma = 2$			$\sigma = 1$		$\sigma = 2$	
γ	s	$s(\eta)$	$s(\hat{\eta})$	s	$s(\eta)$	$s(\hat{\eta})$	s	$s(\eta)$	s	$s(\eta)$
	separable utility, $u = \ln c + \frac{b\left(l^{1-1/\gamma}-1\right)}{1-1/\gamma}$									
0.1	0.023	0.023	0.012	0.099	0.092	0.050	0.020	0.020	0.086	0.080
0.5	0.036	0.035	0.012	0.151	0.140	0.050	0.020	0.020	0.086	0.080
1.0	0.051	0.050	0.012	0.215	0.200	0.050	0.020	0.020	0.086	0.080
2.0	0.081	0.080	0.012	0.344	0.320	0.050	0.020	0.020	0.086	0.080
10.0	0.325	0.320	0.012	1.373	1.280	0.050	0.020	0.020	0.086	0.080
	Cobb-Douglas utility, $u = \frac{\left(c^{\alpha} l^{1-\alpha}\right)^{1-1/\gamma}-1}{1-1/\gamma}$									
0.1	0.067	0.064	0.140	0.322	0.258	0.560	0.058	0.056	0.288	0.224
0.5	0.043	0.042	0.060	0.183	0.168	0.240	0.024	0.024	0.104	0.096
1.0	0.051	0.050	0.050	0.215	0.200	0.200	0.020	0.020	0.086	0.080
2.0	0.073	0.072	0.045	0.308	0.288	0.180	0.018	0.018	0.077	0.072
10.0	0.266	0.262	0.041	1.118	1.050	0.164	0.017	0.016	0.070	0.066

Note: The wage process is $w = 10$ and $\varepsilon \sim N\left(0, \sigma^2\right)$. Furthermore $\alpha = 0.40$, and b is a function of γ so that labor supply equals α when $\sigma^2 = 0$.

The table compares 'true' savings, solved numerically from the first-order conditions, to the amount of savings predicted by the measures of precautionary strength. Note that $\hat{\eta} = \eta$ when labor supply is fixed.

Figure 1
Precautionary strength with Cobb-Douglas utility



The graph relates the precautionary strength for flexible and fixed labor supply (η^{flex} and η^{fix}) to the parameter γ in the utility function

$$u(c,l) = \frac{\left(c^{\alpha} l^{1-\alpha}\right)^{1-1/\gamma} - 1}{1 - 1/\gamma},$$

where $\alpha = 0.40$.